



The impact of doping concentration on structure and photoluminescence of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanocrystals

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ABSTRACT

Cubic Eu^{3+} -doped Lu_2O_3 nanocrystals were synthesized via combined Pechini-foaming method. Doping concentration effect on the structural and luminescence properties was studied. It was found that the increase of Eu^{3+} concentration leads to the growth of unit cell parameters and red shift of Raman lines. The optimal doping concentration and shape of emission lines were studied upon different excitation mechanism. Radiative and nonradiative transition rates and Judd-Ofelt parameters were calculated using the model of 4f–4f intensity theory. The type of multipolar interaction resulting in luminescence concentration quenching was determined.

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1. Introduction

Nanosized materials doped with rare earth ions have attracted extensive attention in the past few decades due to their unique electronic and optical properties and potential applications in the fields of high-performance luminescence devices, magnets, catalysts, medical diagnostics, biological fluorescence labels, etc. [1–5]. Such phosphors often demonstrate significant changes in their properties compared with bulk materials due to the increase of surface/volume ratio and as a result enhancement of surface role [6–8].

Rare earth oxides are widely used as a host lattice for active ion introduction [9–11]. Among them, lutetium oxide attracts increasing attention because of its high density (9.4 g/cm³), high atomic number ($Z=71$), high physical and chemical stability, and a band gap large enough to accommodate the energy levels of many luminescent activator [12]. In addition, Lu_2O_3 has the same cubic structure as Y_2O_3 , a well-known host lattice for efficient, commercial phosphors. It was reported that Eu^{3+} doped lutetia is promising scintillator, especially for medical use [13], therefore it is important to synthesize and study this material.

Nowadays, there are a large variety of approaches to produce

nanosized phosphors doped with rare earth ions with controlled chemical-physical parameters. One of the most cheap and easy way to synthesize nanocrystalline powders is Pechini technique [14]. The main disadvantage of standard Pechini technique is strong agglomeration of derived particles. To overcome this drawback and to improve properties of the obtained nanopowders, we have developed combined Pechini-foaming method applicable for synthesis of simple oxides.

Despite wide application of Eu^{3+} doped Lu_2O_3 particles and numerous papers devoted to the research of this material, to the best of our knowledge, there is a lack of studies dealing with wide concentration series and luminescence quenching in lutetia host.

The aim of this work is to provide the systematic study of doping concentration effect on the structural and luminescence properties of nanocrystalline $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ powders. Analysis of the optimum concentration dependence upon excitation wavelength and determination of the concentration quenching mechanism were also performed.

2. Experimental

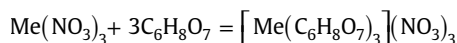
Eu^{3+} doped Lu_2O_3 nanopowders were prepared by the combined Pechini-foaming method. Previously we synthesized complex oxides doped with different rare earth ions using modified Pechini method [15–17]. This method reduces agglomeration but requires two calcination stages.

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The proposed combined technique is also an improved standard Pechini method but it has only one calcination stage. The main part of modification is implementation of foaming agent into polymer gel matrix in order to prevent strong agglomeration of synthesized samples. Unlike $\text{Y}_2\text{O}_3:\text{Eu}^{3+}$ concentration series [18], we used potassium carbonate as a foaming agent.

Solutions of lutetium and europium nitrates were used as precursors. The nitrates had been obtained by dissolving their oxides in concentrated nitric acid. The solutions of the nitrates mixture were heated and then the saturated solution of citric acid was added to the solutions in volume ratio 1:1. The chemical reaction equation can be represented as:



Then potassium carbonate was also added to the solution (weight ratio of the potassium carbonate to the final oxide powder was 1:1). Addition of the ethylene glycol to the metals complexes (volume ratio of the ethylene glycol solution to the citric acid solution was 1:4) leads to the transparent polymer gel formation due to etherification reaction. Pores of the polymer were filled by the potassium carbonate solution. The obtained polymer was annealed at 1000 °C during 90 minutes to remove organic components. The synthesized particles were collected by centrifugation at 2800 rpm for 5 min; repeated 3 times. The resulting washed precipitate was dried in an oven at 110 °C.

The described method was used to synthesize wide concentration series of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ (2–40 at%). The synthesized powder samples (5 mg) and potassium bromide (300 mg) were pressed into pellets (diameter 13 mm) for luminescence studies.

X-ray diffraction patterns were registered with the powder diffractometer UltimaIV (Rigaku) in Bragg-Bretano geometry with $\text{CuK}\alpha 1$ radiation ($\lambda = 1.54059 \text{ \AA}$) in the 2θ range from 10° to 80°. Phase identification was carried out using a powder diffraction database PowderDiffractionFile (PDF-2, 2011). The unit cell parameters were estimated using TOPAS software by full-profile analysis. Electron micrograph images were obtained with Zeiss Supra 40VP scanning electron microscope (resolution 4 nm). Raman spectra were measured with LabRAM with argon laser ($\lambda = 488 \text{ nm}$) as an excitation source. Luminescent spectra were recorded with a fluorescence spectrometer Fluorolog-3 equipped with a Xe-arc lamp (450 W power). For the lifetime measurements the europium luminescence was excited with a Xe-flash lamp (150 W power, 3 μs pulse width). All the measurements were performed at the room temperature.

3. Results and discussion

3.1. Structural analysis

Fig. 1a presents XRD patterns of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanoposphors series with doping concentration $C(\text{Eu}^{3+}) = 2\text{--}40 \text{ at\%}$. All observed peaks in diffraction patterns match well to cubic phase of Lu_2O_3 (JCPDS 43-1021). It can be clearly seen that diffraction lines position significantly shifts towards a lower angle along with doping concentration increase.

The unit cell parameters were calculated with TOPAS software using full-profile analysis. According to calculations, both unit cell parameter and single-crystal cell volume systematically increase with growth of Eu^{3+} ions concentration. This fact can be explained by substitution of lutetium ions ($r = 0.86 \text{ \AA}$) with larger europium ions ($r = 0.95 \text{ \AA}$) [19]. It should be noted that the cell volume has a linear relationship with the content of Eu^{3+} , which is consistent with Vegard's law (Fig. 1b). This result demonstrates that the europium ion has been efficiently and homogeneously incorporated into the host matrix due to the similar ionic radius and chemical reactivity of Eu^{3+} and Lu^{3+} . So, the obtained results testify that the doping ions occupy lutetium sites in crystal lattice.

SEM image of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ 2 at% nanocrystalline powder is shown in Fig. 2. It can be seen, that obtained powder consists of rather small well-faceted nanoparticles. The average size of these nanoparticles was determined to be about 45 nm with standard deviation of 16 nm.

Fig. 3a presents Raman spectra of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanopowders with various doping concentration normalized to the band centered at about 390 cm^{-1} and measured in the spectral region of $80\text{--}700 \text{ cm}^{-1}$.

According to the factor group theory investigation [20], twenty two Raman modes have been predicted for the cubic lutetia lattice belonging to the $Ia\bar{3}$ space group with C type structure.

The major Raman peak is observed at about 391 cm^{-1} and can be assigned to $F_g + A_g$ mode [21]. The high intensity of this band compared to the others indicates a large polarizability variation during the vibration [21]. Other observed lines can be attributed to the following vibration modes: 98 cm^{-1} (F_g), 119 cm^{-1} (F_g), 346 cm^{-1} ($F_g + E_g$), 452 cm^{-1} ($F_g + E_g$), and 611 cm^{-1} ($F_g + A_g$) [21,22]. In our previous manuscript was found out that the line situated at the spectral region $470\text{--}570 \text{ cm}^{-1}$ could not be Raman lines and most likely correspond to the luminescence of Eu^{3+} ions located at positions with C_{3i} symmetry sites [18]. As it can be seen from Fig. 3b, energy levels of Eu^{3+} ions that occupy C_{3i} site are higher than that for C_2 site [23,24]. So, the energy transfer from C_{3i} to the C_2 site occurs, and the transfer efficiency increases with

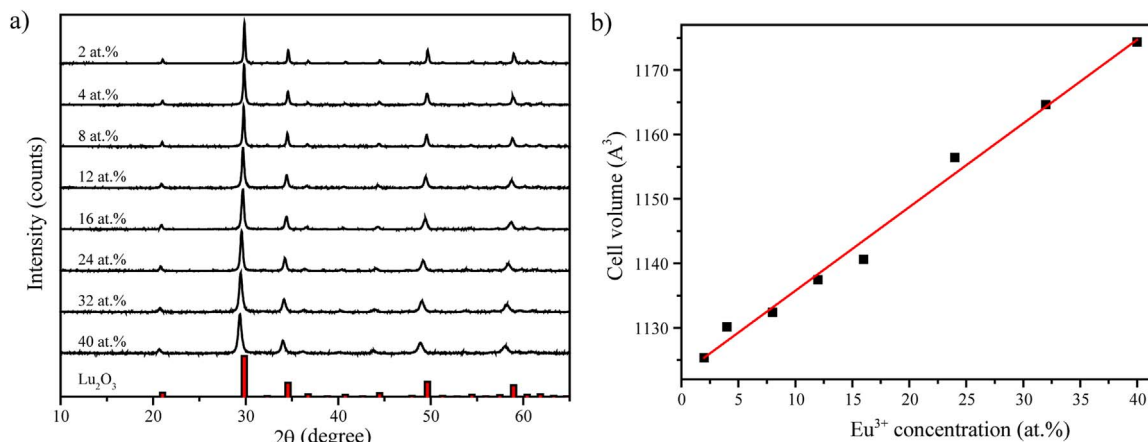


Fig. 1. (a) XRD patterns of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ concentration series and the standard card Lu_2O_3 ; (b) unit cell volume as a function of Eu^{3+} -doping.

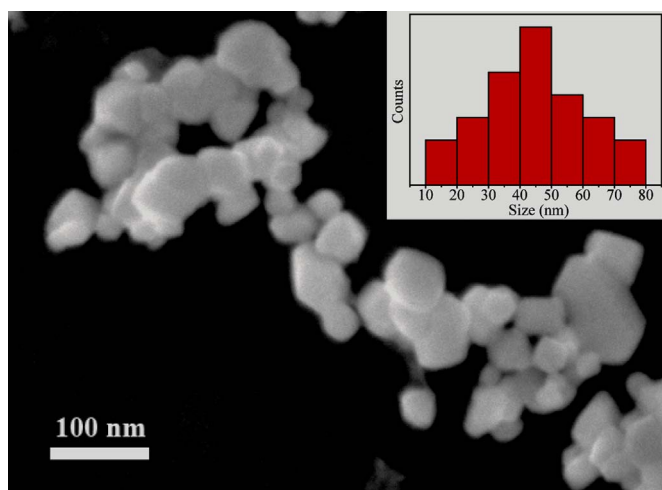


Fig. 2. SEM image of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ 2 at% nanocrystalline powder.

increasing of Eu^{3+} concentration [25,26]. Thus, observed intensity decrease of $470\text{--}570\text{ cm}^{-1}$ band with increasing of doping concentration can be explained by the proposed mechanism of energy transfer.

It should be noted that the spectra do not show the expected number of bands. Some bands are quite broad, and some of them are likely overlapped, at least partially. Probably, nanoscale dimension of samples lead to the inhomogeneous strains and to the band broadening [27]. Furthermore, some Raman bands can overlap with luminescence lines. It was found that doping concentration affects the position of Raman lines. Red shift of peaks was observed along with the increase of Eu^{3+} ions level (inset of Fig. 3). This shift can be explained by changes in the effective mass and effective potential [28,29]. It is well-known that substitution to an element with larger mass and larger electrons number leads to shift toward a lower wavenumber. So, the most intense Raman band was shifted by 12 cm^{-1} for $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ 40 at% sample.

3.2. Photoluminescence properties

Fig. 4a shows the photoluminescence excitation spectrum of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ 12 at% nanophosphor ranging from 240 to 590 nm. This spectrum was monitored for forced electric dipole transition $^5\text{D}_0\text{--}^7\text{F}_2$ at emission wavelength $\lambda_{\text{em}} = 610.6\text{ nm}$. As it can be seen, excitation spectrum consists of broad and strong band in the UV region below 275 nm and series of low intense sharp lines from 300 to 590 nm. The broad band can be assigned to the charge transfer transition (CT) between O^{2-} and Eu^{3+} ions, an electron

transfers from O^{2-} ($2p^6$) orbital to the empty orbital of $4f^6$ for Eu^{3+} . Low intensity peaks are attributed to the typical intra-configurational f-f transitions from ground level $^7\text{F}_0$ to excited levels $^5\text{L}_8$ (322 nm), $^5\text{D}_4$ (362.5 nm), $^5\text{L}_7$ (382 nm), $^5\text{L}_6$ (393 nm), $^5\text{D}_3$ (414 nm), $^5\text{D}_2$ (465 nm), $^5\text{D}_1$ (532.5 nm) and $^5\text{D}_0$ (580 nm). It should be noted that intensity of CT band and f-f transitions differs by two orders of magnitude.

Emission spectrum of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ 12 at% nanophosphor is shown in Fig. 4b. The measurement was carried out in the spectral range 500–750 nm upon 260 nm. Observed spectrum consists of characteristic narrow lines corresponding to the intra-configurational f-f transitions. The luminescence intensity of such transitions strongly depends on the site symmetry of rare earth ions. Eu^{3+} ions can be located at two symmetry sites in cubic Lu_2O_3 host: C_2 (noncentrosymmetric) and C_{3i} (centrosymmetric). According to selection rules, for the Eu^{3+} located at C_{3i} site, only the magnetic dipole transitions can be observed, whereas for the Eu^{3+} ions at C_2 site, the forced electric dipole transitions are also allowed [26].

As can be seen from Fig. 4b, the emission spectrum is dominated by the forced electric dipole transition $^5\text{D}_0\text{--}^7\text{F}_2$ centered at 610.6 nm. Maximum of magnetic dipole transition $^5\text{D}_0\text{--}^7\text{F}_1$ slightly differs for C_2 and C_{3i} sites due to different energy level position. It should be noted that very weak spectral line centered at about 581 nm, which can be assigned to the Eu^{3+} ions located at C_{3i} site, was observed at low-doped samples (2 and 4 at%) only. Three lines centered at 586.4, 592.6 and 599.6 nm correspond to the Eu^{3+} ions located at C_2 site. Other lines in the emission spectrum can be attributed to the following transitions: $^5\text{D}_0\text{--}^7\text{F}_0$ (579.6 nm), $^5\text{D}_0\text{--}^7\text{F}_3$ (650.2, 657.4, 662.8 nm) and $^5\text{D}_0\text{--}^7\text{F}_4$ (686.8, 693.4, 706.8, 708.6, 712.6 nm). Moreover, transition from a higher excited level $^5\text{D}_1$ was also observed: $^5\text{D}_1\text{--}^7\text{F}_1$ (532.4 and 537.4 nm).

Fig. 5 shows concentration dependence of the integrated luminescence intensity of the most intensive transition $^5\text{D}_0\text{--}^7\text{F}_2$ upon excitation wavelengths 260 and 393 nm. It is well-known that the optimum of Eu^{3+} concentration is determined by two competitive effects: on the one hand, an increase of the doping concentration means an increase of luminescence centers number and thus radiative recombination. On the other hand, there is also an increase of the probability of energy transfer between europium ions which enhances the efficiency of the nonradiative processes [23]. As can be seen from Fig. 5, firstly emission intensity goes up with increasing Eu^{3+} concentration but further increase of Eu^{3+} ions number leads to intensity reduction. This indicates the concentration quenching effects for the luminescence of Eu^{3+} in Lu_2O_3 host. It has been reported in earlier studies that there are possibilities of pairing and aggregation of dopant ions at higher doping concentrations where fraction of activators can act as

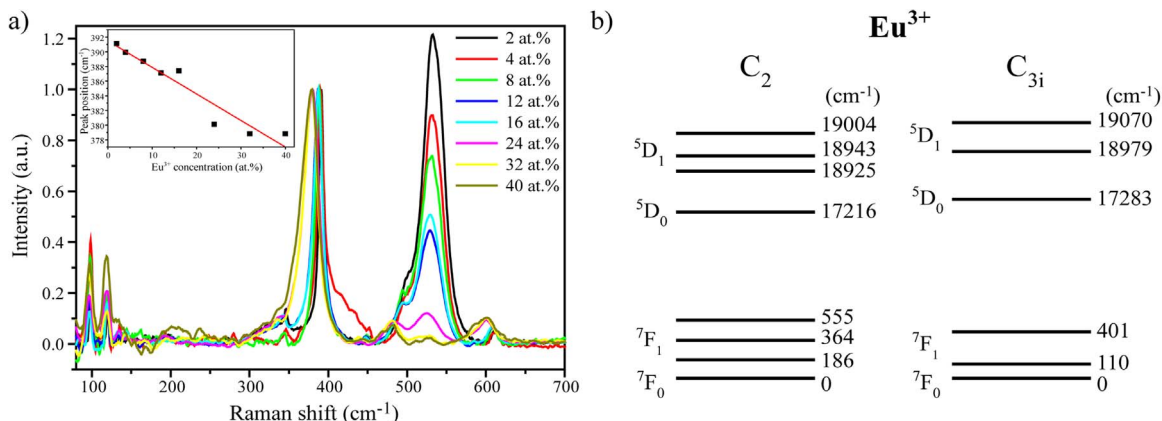


Fig. 3. (a) Raman spectra of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ concentration series, (b) schematic energy levels of Eu^{3+} ions occupied C_2 and C_{3i} symmetry site in Lu_2O_3 host.

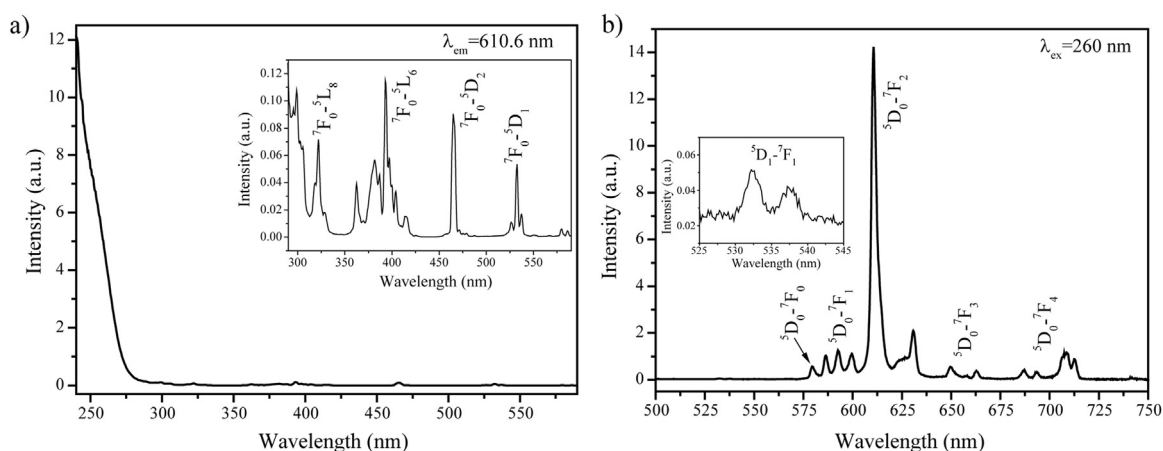


Fig. 4. (a) Excitation and (b) emission spectra of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ 12 at% nanophosphor.

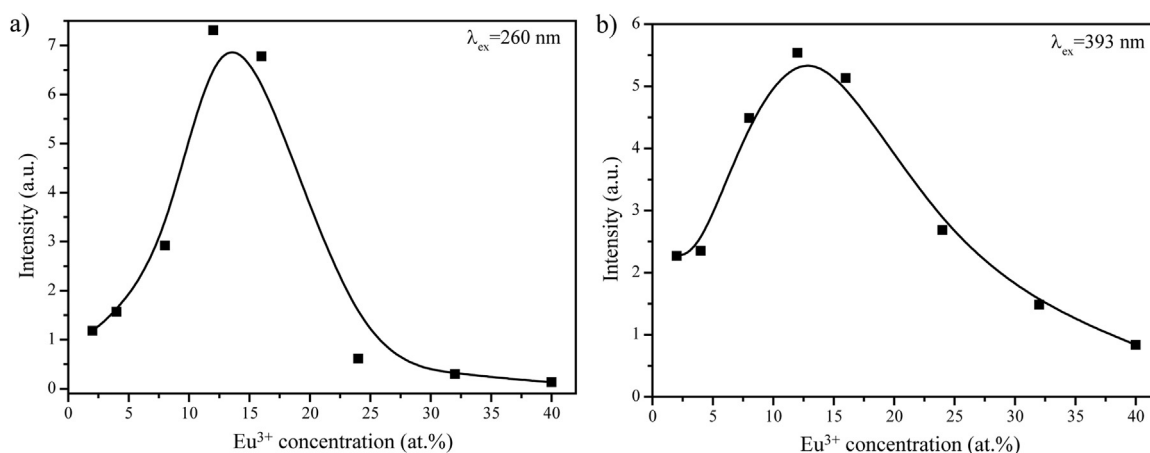


Fig. 5. Photoluminescence intensity as a function of Eu^{3+} doping concentration upon 260 nm (a) and 393 nm (b). Lines are guides for the eye.

quenchers. The optimum doping concentration was determined to be 12 at% for both excitation wavelength. So, it was found that optimum Eu^{3+} concentration in lutetia host does not depend on excitation mechanism. This result differs from what was previously obtained for $\text{MgAl}_2\text{O}_4:\text{Eu}^{3+}$ concentration series, where excitation through charge transfer transition between O^{2-} and Eu^{3+} ions and 4f–4f transition in Eu^{3+} ion led to the different optimum doping concentration [16].

Effect of the excitation wavelength on the emission spectra was also studied. Fig. 6 presents normalized emission spectra of Eu^{3+} -

doped Lu_2O_3 phosphors (2, 12 and 40 at%) upon 260 nm and 393 nm excitation. It should be noted that spectral shape of luminescence lines in Lu_2O_3 host does not depend on excitation mechanism and doping level. Recently, $\text{Y}_2\text{O}_3:\text{Eu}^{3+}$ nanoparticles synthesized by the same combined Pechini-foaming method with mixture of aluminium nitrate and potassium chloride as a foaming agent revealed significant changes of emission spectrum for highly-doped samples upon 322 and 393.5 nm. The observed changes were assigned to the presence of Eu^{3+} ions in a third site completely different from the well-known C_2 or C_{3i} sites [18]. So,

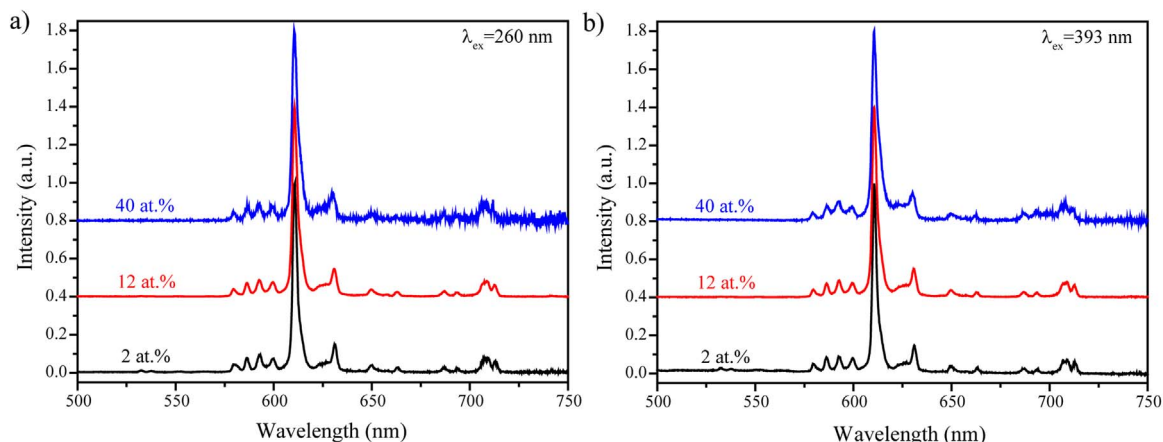


Fig. 6. Normalized emission spectra of Eu^{3+} -doped Lu_2O_3 phosphors upon 260 nm (a) and 393 nm (b).

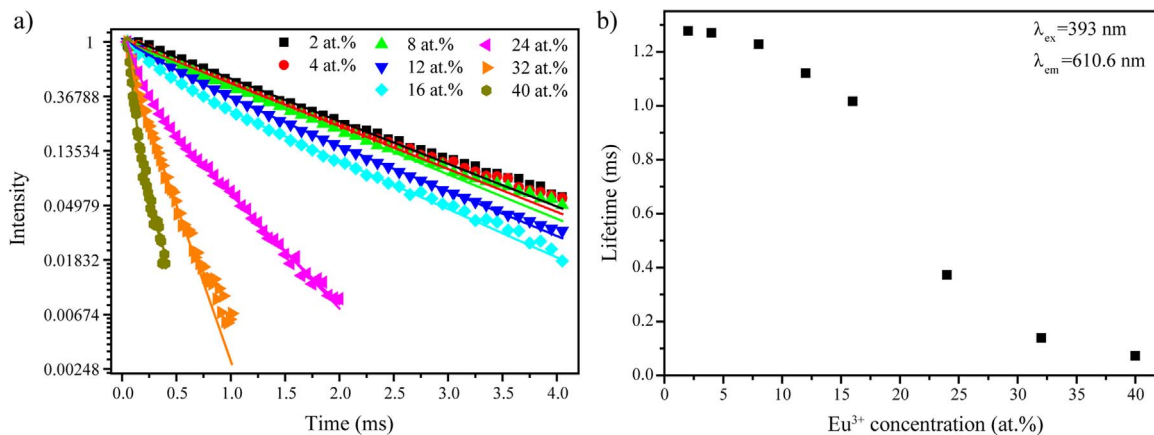


Fig. 7. (a) Decay curves of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanopowders ($\lambda_{\text{ex}}=393$ nm; $\lambda_{\text{em}}=610.6$ nm), (b) $^5\text{D}_0$ level lifetime of Eu^{3+} ions as a function of Eu^{3+} doping level.

we can draw a conclusion that in Lu_2O_3 host Eu^{3+} ions locate only at normal C_2 and C_{3i} sites.

In addition to the steady-state measurements, important information about luminescence properties of synthesized nanopowders could be obtained from fluorescence kinetics. The luminescence decays of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanopowders were measured monitoring forced electric dipole transition intensity $\lambda_{\text{em}}=610.6$ nm (Fig. 7a). The luminescence was excited with 393 nm radiation. Experimental curves of synthesized nanopowders doped up to 8 at% were fitted by single exponential function. However, higher doped samples show non mono-exponential decay, and correct fits required a biexponential model to be used. The average lifetime was calculated by a double exponential fit [30].

$$\tau_{\text{av}} = \frac{A_1\tau_1^2 + A_2\tau_2^2}{A_1\tau_1 + A_2\tau_2} \quad (1)$$

where A_1 and A_2 are pre-exponential factors; τ_1 and τ_2 are lifetimes. The change of fitting model can be explained by the following reason. An increasing concentration of the dopant forces allocation of Eu^{3+} ions in the vicinity of the crystallites surface [31]. So, the non-exponential decay can occur due to the different decays of the Eu^{3+} ions at the surface and Eu^{3+} ions in the volume of the nanoparticles [32–34]. Fig. 7b shows calculated lifetimes as a function of Eu^{3+} doping concentration. As can be seen, the lifetime decreases monotonically from 1.28 ms to 0.07 ms with increasing of Eu^{3+} ions number.

It is well known that lifetime of excited state depends on probabilities of both radiative and nonradiative processes. Radiative decay rate is affected by strength of the dipole transition, determined by the initial and final states; and local-field correction determined by the refractive index of the host matrix. Non-radiative processes include multiphonon relaxation, quenching on impurities (e.g. OH^- group) and cooperative processes (cross-relaxation, energy migration). All these processes were described in detail in our earlier works [35,36]. Briefly, multiphonon relaxation from $^5\text{D}_0$ level is extremely low taking into account energy gap (12000 cm^{-1}) and wavenumber of the most intense vibration (~ 390 cm^{-1}). Probability of quenching on impurities cannot be neglected due to solution-based synthesis method and possibility of OH^- groups to be adsorbed from the atmosphere. From energy level diagram analysis it could be concluded that $^5\text{D}_0$ level is involved in diffusion processes but not in self-quenching processes.

4f–4f intensity theory allows studying the radiative and non-radiative decay rates and quantum efficiencies of phosphors doped with Eu^{3+} ions. These parameters can be calculated from emission spectra, following the method described earlier. In this method

$^5\text{D}_0\text{--}^7\text{F}_1$ transition is taken as an internal standard due to its magnetic dipole character ($A_{0-1}=A_{\text{MD},0}\cdot n_0^3$, where $A_{\text{MD},0}$ is the spontaneous emission probability for the $^5\text{D}_0\text{--}^7\text{F}_1$ transition in vacuo, 14.65 s^{-1}) [37]. Dealing with nanopowders, we should substitute usual refractive index n_0 to effective refractive index n_{eff} , which consists of the Lu_2O_3 nanoparticles surrounded by the media with refractive index n_{med} (in our case surrounding media is potassium bromide) [8,38].

$$n_{\text{eff}} = x \cdot n_{\text{Lu}_2\text{O}_3} + (1-x) \cdot n_{\text{med}} \quad (2)$$

where x is the “filling factor” showing what fraction of space is occupied by the nanoparticles.

The Einstein coefficients $A_{0-\lambda}$ for spontaneous emission can be obtained from the relationship:

$$A_{0-\lambda} = A_{0-1} \frac{\nu_{0-1} I_{0-\lambda}}{\nu_{0-\lambda} I_{0-1}} \quad (3)$$

where $\nu_{0-\lambda}$ and $I_{0-\lambda}$ are, respectively, the frequency and intensity of the corresponding transition $^5\text{D}_0\text{--}^7\text{F}_\lambda$ in the emission spectrum. The radiative probability is sum of all the $A_{0-\lambda}$ values. Using observed lifetime and calculated radiative decay rate nonradiative probability and quantum efficiency also can be determined.

The Judd–Ofelt intensity parameters (Ω_2 and Ω_4) can be used to study nature of the luminescence behaviour of Eu^{3+} ions. The Einstein coefficient of the spontaneous emission is expressed as:

$$A_{0-\lambda} = \frac{64\pi^4 \nu^3 e^2}{3hc^3} \frac{1}{4\pi\epsilon_0} \chi \sum_{\lambda=2,4} \Omega_\lambda \left| \langle ^5\text{D}_0 \| U^\lambda \| ^7\text{F}_j \rangle \right|^2 \quad (4)$$

where $\chi = n_{\text{eff}}^4(n_{\text{eff}}^2 + 2)^2/9$ is Lorenz local field correction and $n_{\text{eff}}=1.79$ is the effective index of refraction of the medium. The square reduced matrix $\left| \langle ^5\text{D}_0 \| U^{(\lambda)} \| ^7\text{F}_2 \rangle \right|^2 = 0.0032$ and $\left| \langle ^5\text{D}_0 \| U^{(\lambda)} \| ^7\text{F}_4 \rangle \right|^2 = 0.0023$ were taken from [39].

The calculated values of radiative and nonradiative decay rates, quantum efficiencies, and Judd–Ofelt intensity parameters are presented in Table 1.

As can be seen from Table, radiative decay rate almost is not changed in concentration series. It means that Eu^{3+} introduction in lutetia host only slightly affects strength of the dipole transition and local-field correction. Nonradiative decay rate monotonically increased with doping concentration; however, drastically growth was observed only for highly-doped samples. Among all types of nonradiative processes only cooperative processes are strongly affected by doping level. Thus, the observed growth of non-radiative decay rate can be explained by the increase of spatial energy migration with subsequent quenching on impurities.

Table 1

Radiative (A_{rad}), nonradiative (A_{nrad}) and total (A_{total}) transition rates of the 5D_0 level, quantum efficiencies (η) and Judd–Ofelt intensity parameters (Ω_2 and Ω_4) as a function of Eu^{3+} doping concentration in Lu_2O_3 host.

Eu^{3+} concentration (at%)	$A_{\text{rad}} (\text{s}^{-1})$	$A_{\text{nrad}} (\text{s}^{-1})$	$A_{\text{total}} (\text{s}^{-1})$	$\eta (\%)$	$\Omega_2 (10^{-19} \text{ cm}^2)$	$\Omega_4 (10^{-19} \text{ cm}^2)$
2	590	190	780	76	7.8	2.9
4	670	120	790	85	8.8	3.6
8	690	130	820	84	9.1	3.6
12	690	200	900	78	9.2	3.7
16	690	290	990	71	9.2	3.7
24	690	2000	2700	26	9.0	3.8
32	670	6500	7200	9	8.6	4.2
40	660	13,000	13,700	5	8.2	4.5

Similar phenomenon was previously observed for different oxide hosts doped with Eu^{3+} ions [16,35,36].

It is well-known that Ω_2 parameter reflects environment around the rare earth ion and covalency of RE–O bond, whereas Ω_4 parameter is affected by electron density on the surrounded O^{2-} ions. It should be noted that values of Ω_2 and Ω_4 intensity parameters are quite high for all studied samples. Ω_2 intensity parameter demonstrated quite small changes, but Ω_4 intensity parameter gradually increased with growth of doping concentration. The observed growth can be explained by decrease of the electron density on O^{2-} ions [40].

As it was found out for highly-doped samples, luminescence intensity reduces along with increase of Eu^{3+} concentration due to the excitation migration with subsequent quenching on the impurities. Blasse [41] has pointed out that if the activator was introduced solely on one crystallographic site, the mechanism of energy transfer in phosphors can be discussed in terms of the critical energy transfer distance (R_c). Although, there are two different crystallographic sites for Eu^{3+} ions in Lu_2O_3 host, no luminescence signal from Eu^{3+} ions located at C_{3i} site was observed for samples with doping concentration higher than 4 at% due to the energy transfer from C_{3i} to C_2 site. Therefore, we can consider only Eu^{3+} situated at C_2 sites. The critical energy transfer distance was calculated by the following equation:

$$R_c \approx 2 \left[\frac{3V}{4\pi x_c N} \right]^{1/3} \quad (5)$$

where x_c is the critical doping content ($x_c = 0.12$), N is the number of cation sites in the unit cell ($N = 24$ for C_2 sites in Lu_2O_3), and V is the volume of the unit cell ($V \approx 1137.44 \text{ \AA}^3$). Using these parameters, R_c was determined to be 9 Å. Since R_c is quite big (not less than 5 Å), exchange interaction does not play considerable role in nonradiative energy transfer process from one Eu^{3+} ion situated in C_2 site to another in Lu_2O_3 host [42]. Therefore, observed concentration quenching is explained by the multipolar interaction mechanism [43,44]. Multipolar interaction involves several types of interaction such as dipole–dipole (d–d), dipole–quadrupole (d–q), and quadrupole–quadrupole (q–q) interaction. In order to determine interaction type between sensitizers or between sensitizer and activator, Van Uitert [45] proposed formula that was modified to a more practical form by Ozawa and Jaffe as follows [46]:

$$\frac{I}{x} = k \left[1 + \beta(x)^{\theta/3} \right]^{-1} \quad (6)$$

where I is the integral intensity, x is the activator concentration, k and β are constant for the same excitation conditions for a given host crystal. According to the above equation $\theta = 3$ for the energy transfer among the nearest neighbor ions, while $\theta = 6, 8$ and 10 for d–d, d–q and q–q interactions, respectively [47,48].

Since the critical concentration of Eu^{3+} ions was determined as

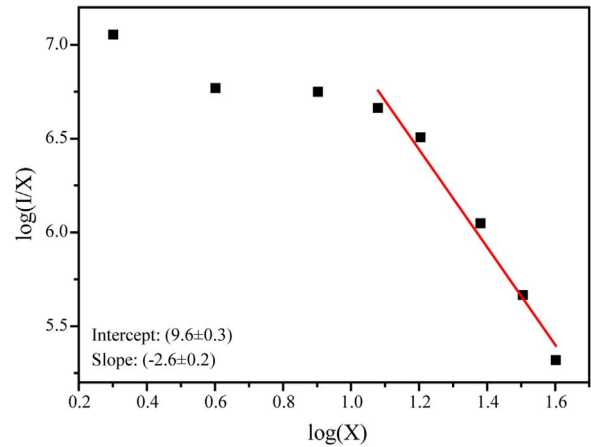


Fig. 8. Relation between $\log(I/x)$ and $\log(x)$ in $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanopowders. Line is a linear fit.

12%, the dependence of the emission intensity of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanopowders excited at 393 nm as a function of the doping concentration greater than the critical concentration was studied. The multipolar character (θ) can be obtained by plot $\log(I/x)$ vs $\log(x)$ as presented in Fig. 8. The slope $\theta/3$ from linear fitting function was found to be -2.6 . The value of θ can be calculated as 7.8, which is close to 8. Therefore, concentration quenching in $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ phosphor occurred due to dipole–quadrupole (d–q) interaction.

4. Conclusion

Concentration series of $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ nanopowders were synthesized by combined Pechini-foaming technique with K_2CO_3 as a foaming agent. XRD patterns indicated formation of pure cubic phase without any impurity. SEM images showed that the synthesized nanocrystalline powders consist of rather small well-faceted nanoparticles with average size about 50 nm. The strongest Raman line corresponded to $F_g + A_g$ mode and was centered at about 390 cm^{-1} . It was observed that the luminescence intensity of Eu^{3+} ions situated at C_{3i} site reduced with increase of doping concentration. Emission spectrum consisted of characteristic narrow lines corresponding to the intra-configurational f–f transitions and is dominated by forced electric dipole $^5D_0 \rightarrow ^7F_2$ transition. Optimal Eu^{3+} doping concentration was determined to be 12 at% in Lu_2O_3 host regardless excitation mechanism. Observed lifetime of 5D_0 level was gradually decline from 1.28 ms to 0.07 ms along with growth of Eu^{3+} content. Radiative and nonradiative transition rates and Judd–Ofelt parameters were calculated using the model of 4f–4f intensity theory. It was found that concentration quenching in $\text{Lu}_2\text{O}_3:\text{Eu}^{3+}$ phosphor occurred due to dipole–

quadrupole energy migration with subsequent quenching on impurities.

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